In some situations, change is kept in check by the loop. For example, if aphids suddenly multiply, they provide more food for ladybugs, leading to an increase in the number of ladybugs. But with more ladybugs feeding on the aphids, aphid numbers drop again. This is negative feedback and it helps maintain the status quo.

In other cases, feedback can accelerate change. Shrubs, for example, may begin to take over from grass on newly colonized land, casting their shade over the grass, depriving it of sunlight, and slowing its growth. The shrubs now have more water and nutrients, so they prosper at the expense of the grass. This is positive feedback and it is inherently destabilizing.

Ideas about feedback loops first developed early in the 20th century. They were based on the work of two mathematicians—Alfred Lotka (1880–1949) in the US and Italian Vito Volterra (1860–1940)—who independently devised equations based on the interaction between predators and prey. Their equations showed that a prey population will grow rapidly when the number of predators drops, while the predator population will drop when prey numbers drop, because the predators go hungry. The result is a constant cycle of falling and rising predator and prey populations.